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Demo Of 555 Timer-Based ASTABLE MULTIVIBRATOR Using MATLAB

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In analogue electronics, oscillators and their implementation using integrated circuits (ICs) is an important subject. As 555 timer IC is easy to understand and flexible enough to suit diverse applications, it is used in astable multivibrator (oscillator) configuration for study purpose. While covering the topic in the classroom, instructors often use circuit simulators such as Proteus and TINA to show the output waveform variation of 555 timer IC by changing the resistor or capacitor values.

We present here a demo program for a 555 timer-based astable multivibrator, which is implemented using the graphical user interface (GUI) in MATLAB 2014 environment, as shown in Fig. 1. Two resistors and a capacitor are required to operate 555 timer IC in astable mode. When the user enters values of the external components (R1, R2 and C) and presses 'Run' button, the simulator provides the time period, frequency and duty cycle of the square wave. The program also shows the output square waveform in the figure window (Fig. 2). The GUI also shows the circuit diagram for operating timer 555 in astable mode.

Software

As mentioned before, 555 timer IC is used in astable mode in order to produce a square wave. Time period (or frequency) and duty cycle of the astable multivibrator is determined by external components R1, R2 and C (refer to the circuit diagram in Fig. 1).

The time during which the output is high:

$T_h = 0.693 \times (R1 + R2) \times C$ seconds

The time during which the output is low:

$T_l = 0.693 \times R2 \times C$ seconds

Therefore the time period of the square wave is:

$T = T_h + T_l = 0.693 \times \{R1 + (2 \times R2)\} \times C$ seconds

and frequency in Hertz (Hz) is:

$f = 1/T = 1.44/\{R1 + (2 \times R2)\} \times C$

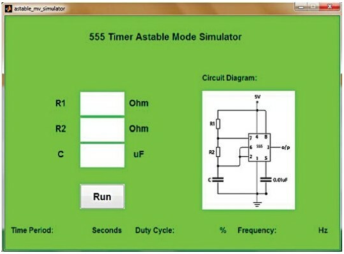

Fig. 1: The GUI for the 555 timer astable mode simulator

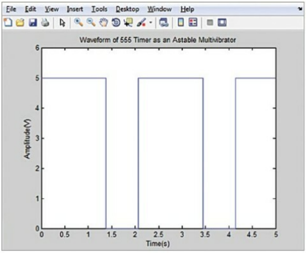

Fig. 2: Waveform for R1=1000 ohms, R2=1000 ohms and C=1000µF